



Shore

Year 12

HSC Assessment Task 5 - Trial HSC

17th August 2012

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this question paper
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- Start each of Questions 11–14 in a new writing booklet
- Show all necessary working in Questions 11–14
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

Total marks – 70

Section I Pages 3–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–12

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1** The point P divides the interval AB , where $A = (-8, -2)$ and $B = (16, 10)$, internally in the ratio $1 : 3$. What are the coordinates of P ?

(A) $(-2, 1)$

(B) $(0, 2)$

(C) $(8, 6)$

(D) $(10, 7)$

- 2** The equation $x^3 - 5x + 2 = 0$ has roots α , β and 2. What is the value of $\alpha + \beta$?

(A) -7

(B) -2

(C) 2

(D) 3

- 3** What is the coefficient of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^5$?

(A) -160

(B) -80

(C) -32

(D) -2

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

4 What is the exact value of $\int_{-4}^4 \frac{dx}{x^2 + 16}$?

- (A) 0
- (B) $\frac{\pi}{16}$
- (C) $\frac{\pi}{8}$
- (D) $2 \ln 32$

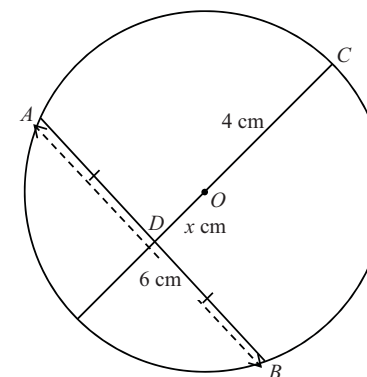
5 Suppose $x^3 - (a+1)x + a \equiv (x+3)Q(x)$ where $Q(x)$ is a polynomial. Find the value of a .

- (A) -6
- (B) -3
- (C) 3
- (D) 6

6 The curves $y = x^2$ and $y = x^3$ intersect at the point (1, 1). Which of the following is closest to the size in radians of the acute angle between these curves at (1, 1)?

- (A) 0
- (B) 0.14
- (C) 0.62
- (D) $\frac{\pi}{4}$

7 In the diagram below the length of chord AB is 6 cm, $AD = BD$, the length of the radius OC is 4 cm and the length of OD is x cm.



NOT TO
SCALE

What is the value of x ?

- (A) 2 cm
- (B) $\sqrt{5}$ cm
- (C) $\sqrt{7}$ cm
- (D) 5 cm

8 It is known that $x^3 + 3x - 7 = 0$ has a root between $x = 1$ and $x = 2$. If the method of halving the interval is used twice, between which two values does the root lie?

- (A) $x = 1$ and $x = 1.25$
- (B) $x = 1.25$ and $x = 1.5$
- (C) $x = 1.5$ and $x = 1.75$
- (D) $x = 1.75$ and $x = 2$

9 Which one of these functions has an inverse relation that is not a function?

- (A) $y = x^3$
- (B) $y = \ln x$
- (C) $y = \sqrt{x}$
- (D) $y = |x|$

10 Which one of the following is the general solution of $2\sin^2\left(6t + \frac{\pi}{4}\right) = 1$?

- (A) $t = \frac{n\pi}{12}$, where n is an integer.
- (B) $t = \frac{n\pi}{12} - \frac{\pi}{24}$, where n is an integer.
- (C) $t = \frac{n\pi}{3}$ and $t = \frac{n\pi}{3} + \frac{\pi}{12}$, where n is an integer.
- (D) $t = \frac{n\pi}{3} - \frac{\pi}{6}$ and $t = \frac{n\pi}{3} + \frac{\pi}{12}$, where n is an integer.

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Start each of Questions 11–14 in a new writing booklet.

Question 11 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) Find the exact value of $\cos\left(\sin^{-1}\frac{4}{7}\right)$. **2**

(b) (i) Graph $y = |3x - 4|$. **1**

(ii) Hence, or otherwise, solve $|x - 6| \leq |3x - 4|$. **2**

(c) Find $\frac{d}{dx}(x \cos^{-1} 2x)$. **2**

(d) Use the substitution $u = x^3 + 1$ to find $\int \frac{x^2}{\sqrt{x^3 + 1}} dx$. **3**

(e) By using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that **2**

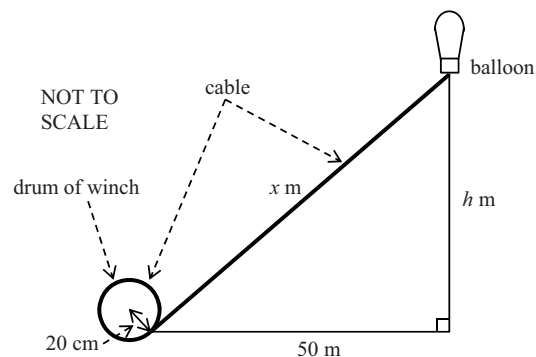
$$\frac{\sqrt{1 - \sin \theta}}{\sqrt{1 + \sin \theta}} = \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}.$$

(f) Find the exact value of the volume of the solid of revolution formed when the region bounded by the curve $y = 3 \cos \frac{x}{4}$, the x -axis and the lines $x = 0$ and $x = \pi$ is rotated about the x -axis. **3**

Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) (i) How many nine-letter arrangements of the letters in the word SKEDADDLE are possible? 1
- (ii) How many nine-letter arrangements of the letters in the word SKEDADDLE are possible if the letters S and K must be adjacent? 1
- (b) A balloon ride consists of a hot-air balloon tethered to a winch by a strong thin cable. The drum of the winch has a radius of 20 cm. The operator of the ride must keep the balloon directly above its landing area, which is 50 m downwind of the winch. Let h metres be the height of the hot-air balloon and x metres be the length of the cable between the balloon and where it meets the winch.

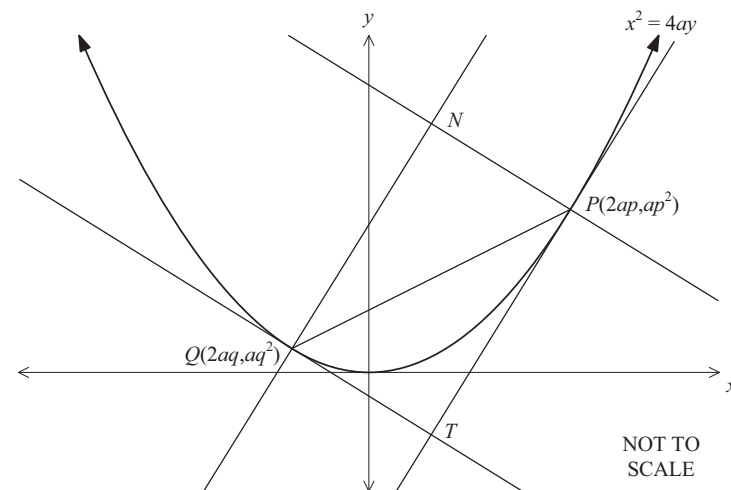


- (i) Find an expression for x in terms of h . 1
- (ii) The winch operator must turn the winch so that the balloon rises vertically at a rate of 0.5 metres per second.
- At what rate $\frac{d\theta}{dt}$ must the winch operator turn the winch when the balloon is at a height of 20 metres? 3
- (c) Sketch the graph of the function $f(x) = 3 \cos^{-1}(2x - 1)$. Clearly indicate the domain and range of the function. 2

Question 12 continues on the following page

Question 12 (continued)

- (d) The diagram shows the parabola $x^2 = 4ay$. The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola such that PQ is a focal chord. The tangents at P and Q intersect at T and the normals at P and Q intersect at N .



- (i) Show that the equation of any chord PQ is given by $y = \left(\frac{p+q}{2}\right)x - apq$. 2
- (ii) Show that $pq = -1$ when PQ is a focal chord. 1
- (iii) Explain why $PTQN$ is a cyclic quadrilateral. 1
- (iv) Let C be the centre of the circle $PTQN$. Explain why C is the midpoint of PQ . 1
- (v) Find the Cartesian equation of the locus of C . 2

Question 13 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) Use mathematical induction to prove that for all integers $n \geq 1$, $11^{n+1} + 12^{2n-1}$ is divisible by 133. 3
- (b) A particle moves in a straight line along the x -axis so that its acceleration is given by $\ddot{x} = x + 3$ where x is the displacement from the origin.

Initially the particle is at the origin and has velocity $v = 3$.

- (i) Show that $v = x + 3$. 2
- (ii) Find x as a function of t . 2
- (c) A multiple choice test consists of 30 questions, each having 4 possible answers. Ferdinand decides to answer each question by randomly choosing one of the four possible answers.
- (i) What is the probability that Ferdinand will answer exactly half of the questions correctly? 1
- (ii) Find the most likely number of questions that Ferdinand answers correctly and the probability that he answers this many questions correctly. Give your answer correct to three significant figures. 3
- (d) A hot frying pan is cooling in a room of constant temperature 20°C . At time t minutes its temperature T decreases according to the equation

$$\frac{dT}{dt} = -k(T - 20) \text{ where } k \text{ is a positive constant.}$$

The initial temperature of the frying pan is 160°C and it cools to 100°C after 10 minutes.

- (i) Verify that $T = 20 + Ae^{-kt}$ is a solution of this equation, where A is a constant. 1
- (ii) Find the values of A and k . 2
- (iii) How long will it take for the pan to become cool enough to touch (50°C)? Give your answer correct to the nearest minute. 1

Question 14 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) A particle moves in a straight line. Its displacement, x metres, after t seconds is given by

$$x = \frac{2}{\sqrt{3}} \cos(3t) + 2 \sin(3t).$$

- (i) Show that the particle is moving in simple harmonic motion by showing that $\ddot{x} = -n^2x$. 2
- (ii) By expressing x in the form $R \sin(3t + \alpha)$, with $0 < \alpha < \frac{\pi}{2}$, find the period and amplitude of the motion. 2

- (b) The binomial theorem states that 3

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k.$$

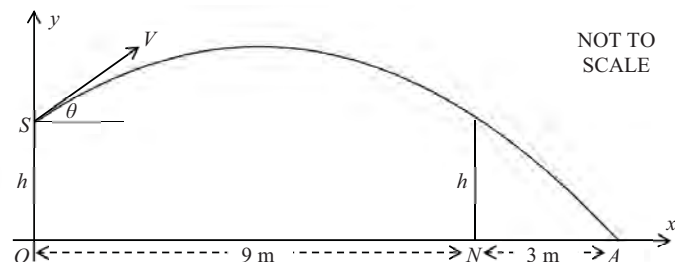
By integrating both sides of this identity with respect to x , show that

$$\frac{2^{n+1} - 1}{n + 1} = \sum_{k=0}^n \frac{1}{k + 1} \binom{n}{k}.$$

Question 14 continues on the following page

- (c) In a game of volleyball the server, S , stands 9 metres from the net, N , and attempts to land a serve on the attack line, A , which is 3 metres from the net on the opposite side of the court.

At time $t = 0$, the ball is hit from S at a speed of V metres per second and an angle of elevation θ . Both the release point of the serve and the height of the net are h metres above the surface of the court. The point directly below S on the surface of the court should be taken as the origin, O , of the coordinate system.



The horizontal displacement of the ball at time t is given by $x = Vt \cos \theta$.
(Do NOT show this).

- (i) The equation of motion of the ball in the vertical direction is $\ddot{y} = -g$. 2
Using calculus, show that the vertical displacement of the ball at time t is given by

$$y = Vt \sin \theta - \frac{1}{2}gt^2 + h.$$

- (ii) Hence show that the trajectory of the ball is given by 1

$$y = h + x \tan \theta - \frac{g}{2V^2 \cos^2 \theta} x^2.$$

- (iii) Show that when the ball clears the net 2

$$V^2 > \frac{9g}{2 \sin \theta \cos \theta}.$$

- (iv) After clearing the net the ball hits the attack line A . Show that 2

$$\tan \theta > \frac{h}{4}.$$

- (v) If $h = 2.43$, find the minimum value of θ at which the ball can be served so that it clears the net and hits the attack line. 1

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

Section 1

1. A
2. B
3. B
4. C
5. D
6. B
7. C
8. B
9. D
10. A

$$4. \int_{-\pi}^{\pi} \frac{dx}{x^2+4} = \left[\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right)$$

$$= \frac{\pi}{2}$$

$$5. P(x) = x^3 - (a+1)x + a$$

$$P(-3) = 0$$

$$-27 - 3(a+1) + a = 0$$

$$4a + 3 = 27$$

$$a = 6$$

$$6. y = x^2 \text{ at } (1,1) \Rightarrow m_1 = 2$$

$$y = x^3 \text{ at } (1,1) \Rightarrow m_2 = 3$$

$$\therefore \tan \theta = \left| \frac{2-3}{1+2 \times 3} \right|$$

$$\theta = \tan^{-1}\left(\frac{1}{7}\right)$$

$$\approx 0.14$$

Section 1 - Worked.

$$1. P = \left(\frac{3x-8+1 \times 16}{1+3}, \frac{3x-2+1 \times 10}{1+3} \right)$$

$$= (-2, 1)$$

$$2. \alpha + \beta + 2 = 0$$

$$\therefore \alpha + \beta = -2$$

$$3. T_{k+1} = \binom{n}{k} x^{n-2k} \left(-\frac{2}{x}\right)^k$$

$$= \binom{5}{k} x^{10-2k} (-2)^k (x^{-1})^k$$

$$\therefore 10 - 3k = 1$$

$$k = 3$$

$$\therefore \text{coefficient of } x$$

$$= \binom{5}{3} \times (-2)^3$$

$$= -80$$

$$7. \begin{array}{c} D \\ \diagup \quad \diagdown \\ 3 \quad 4 \end{array}$$

$$x = \sqrt{4^2 - 3^2}$$

$$= \sqrt{7}$$

$$8. f(1) = -3, f(2) = 7$$

$$f(1.5) = \frac{2}{3}, f(1.25) < 0$$

$$\therefore 1.25 < x \leq 1.5$$

$$9. y = |x|$$

$$10. \sin\left(6t + \frac{\pi}{4}\right) = \pm \frac{1}{\sqrt{2}}$$

$$6t + \frac{\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}, \dots$$

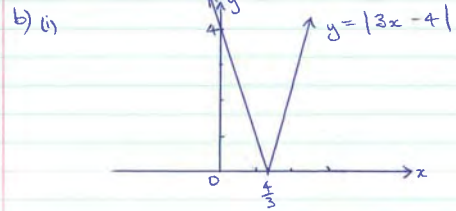
$$6t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}, \dots$$

$$t = 0, \frac{\pi}{12}, \frac{2\pi}{12}, \frac{3\pi}{12}, \frac{4\pi}{12}, \dots$$

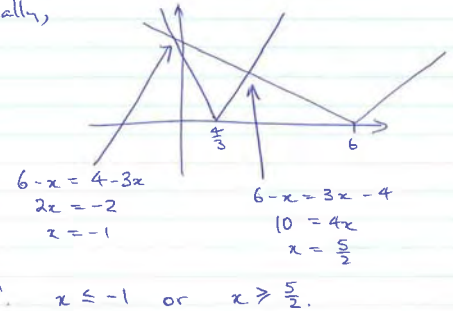
$$= \frac{n\pi}{12}, n \in \mathbb{Z}$$

Section 2 - Question 11.

$$a) \cos\left(\sin^{-1}\left(\frac{4}{7}\right)\right) = \frac{\sqrt{33}}{7}$$



(ii) Graphically,



Question 11

$$c) \frac{d}{dx} (x \cos^{-1} 2x) = \frac{d(x)}{dx} \cdot \cos^{-1} 2x + x \cdot \frac{d(\cos^{-1} 2x)}{dx}$$

$$= \cos^{-1} 2x - \frac{x}{\sqrt{\frac{1}{4} - x^2}}$$

$$\text{or } = \cos^{-1} 2x - \frac{2x}{\sqrt{1 - 4x^2}}$$

$$d) \int \frac{x^2}{\sqrt{x^3+1}} dx$$

$$= \frac{1}{3} \int \frac{1}{\sqrt{u}} du$$

$$= \frac{1}{3} \cdot 2u^{\frac{1}{2}} + C$$

$$= \frac{2}{3} \sqrt{x^3+1} + C$$

$$e) \text{LHS} = \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}}$$

$$= \sqrt{\frac{1 - \frac{2t}{1+t^2}}{1 + \frac{2t}{1+t^2}}}$$

$$= \sqrt{\frac{1+t^2-2t}{1+t^2+2t}}$$

$$= \sqrt{\frac{(1-t)^2}{(1+t)^2}}$$

$$= \frac{1-t}{1+t}$$

$$= \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}$$

$$= \text{RHS}$$

Question 11

$$A) V = \pi \int_0^{\pi} (3 \cos \frac{x}{4})^2 dx$$

$$= 9\pi \int_0^{\pi} \cos^2(\frac{x}{4}) dx$$

$$= 9\pi \int_0^{\pi} \frac{\cos \frac{x}{2} + 1}{2} dx$$

$$= 9\pi \left[\sin(\frac{x}{2}) + \frac{x}{2} \right]_0^{\pi}$$

$$= 9\pi (1 + \frac{\pi}{2} - 0 - 0)$$

$$= 9\pi (1 + \frac{\pi}{2})$$

Question 12

$$a) \text{(i) arrangements} = \frac{9!}{3!2!}$$

$$= 30240$$

$$\text{(ii) arrangements} = 2 \times 8 \times \frac{7!}{3!2!}$$

$$= 6720$$

$$b) \text{(i) } x^2 = h^2 + 50^2$$

$$x = \sqrt{h^2 + 2500}$$

$$\text{(ii) } \frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dh} \times \frac{dh}{dt}$$

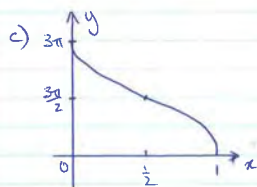
$$= 5 \times 1 \times \frac{h}{\sqrt{h^2 + 2500}} \times 0.5$$

$$= \frac{2.5}{\sqrt{2900}} \times 20$$

$$\approx 0.93 \text{ radians per second}$$

$$\left. \begin{array}{l} l = r\theta \\ \frac{dl}{dt} = r \\ \frac{d\theta}{dt} = \frac{1}{r} \\ = \frac{1}{0.2} \\ = 5 \\ \frac{dl}{dx} = 1 \end{array} \right\}$$

2012 HSC Trial - Maths Extension 1 Solutions



$$0 \leq x \leq 1 \quad \text{Domain}$$

$$0 \leq y \leq 3 \quad \text{Range}$$

d) i) $m_{PA} = \frac{y_P - y_A}{x_P - x_A} = \frac{ap^2 - aq^2}{2ap - 2aq} = \frac{a(p-q)(p+q)}{2a(p-q)} = \frac{p+q}{2}$

PA: $y - y_P = m_{PA}(x - x_P)$
 $y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$
 $y = \left(\frac{p+q}{2}\right)x - \frac{2ap^2}{2} - \frac{2apq}{2} + \frac{2ap^2}{2}$
 $y = \left(\frac{p+q}{2}\right)x - apq$

ii) For focal chord, $(0, a)$ lies on PA.

$$\therefore a = \left(\frac{p+q}{2}\right) \cdot 0 - apq$$

$$a = -apq$$

$$pq = -1$$

iii) $\angle NPT = 90^\circ$ ($NP \perp PT$, given)
 $\angle NQT = 90^\circ$ ($NQ \perp QT$, given)
 $\therefore$ opposite angles of PTQN add to 180° .
 $\therefore$ PTQN is a cyclic quadrilateral.

iv) PT has slope p , QT has slope q .
 And $pq = -1$.
 $\therefore PT \perp QT$.
 $\therefore \angle QTP = 90^\circ$.
 $\therefore$ AP is a diameter of PTQN (angle in a semicircle).
 $\therefore$ the midpoint of PQ is the centre of C.

Question 12

d) (i) $C = \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2}\right)$
 $x = a(p+q) \Rightarrow p+q = \frac{x}{a}$
 $y = \frac{a}{2}(p^2+q^2)$
 $= \frac{a}{2}[(p+q)^2 - 2pq]$
 $= \frac{a}{2}\left(\left(\frac{x}{a}\right)^2 + 2\right)$ as $pq = -1$.
 $= \frac{a}{2}\left(\frac{x^2}{a^2} + 2\right)$
 $y = \frac{x^2}{2a} + a$ or $2a(y-a) = x^2$

Question 13

a) Prove true for $n=1$
 $11^2 + 12^2 = 121 + 144 = 265$ which is divisible by 133.

Assume true for $n=k$.

i.e. $11^{k+1} + 12^{2k+1} = 133m$, where m is an integer.

Prove true for $n=k+1$.

$\therefore$ R.T.P. $11^{k+2} + 12^{2k+2} = 133n$, where n is an integer.

$$11^{k+2} + 12^{2k+2} = 11 \times 11^{k+1} + 144 \times 12^{2k+1}$$

$$= 11 \times (133m - 12^{2k+1}) + 144 \times 12^{2k+1}$$

by the inductive hypothesis

$$= 11 \times 133m + 12^{2k+1}(144 - 11)$$

$$= 133(11m + 12^{2k+1})$$

$$= 133n, \text{ where } n \text{ is an integer.}$$

$\therefore$ by the principle of mathematical induction, true for $n \geq 1$.

2012 HSC Trials - Maths Extension 1 Solutions

Question 13

(b) i) $\ddot{x} = x+3$

$$\frac{d(\frac{1}{2}v^2)}{dx} = x+3$$

$$\frac{1}{2}v^2 = \int (x+3) dx$$

$$= \frac{x^2}{2} + 3x + C_1$$

$$v^2 = x^2 + 6x + C_2$$

When $x=0, v=3$.

$$9 = 0 + 0 + C_2$$

$$C_2 = 9$$

$$\therefore v^2 = x^2 + 6x + 9$$

$$v^2 = (x+3)^2$$

$$v = \pm(x+3)$$

But $v=3$ when $x=0$.

$$\therefore v = x+3$$

ii) $\frac{dx}{dt} = x+3$

$$\frac{dx}{x+3} = \frac{1}{x+3}$$

$$t = \ln(x+3) + C$$

When $t=0, x=0$.

$$0 = \ln 3 + C$$

$$\therefore C = -\ln 3$$

$$t = \ln(x+3) - \ln 3$$

$$e^t = \frac{x+3}{3}$$

$$x = 3e^t - 3$$

2012 HSC Trials - Maths Extension 1 Solutions

Question 13

c) i) $P(\text{correct choice}) = \frac{1}{4}$

$$\therefore P(15 \text{ correct choices}) = \binom{30}{15} \left(\frac{1}{4}\right)^{15} \left(\frac{3}{4}\right)^{15}$$

$$= 0.001930545 \dots$$

ii) Let P_k be $P(k \text{ correct choices})$.

$$\frac{P_{k+1}}{P_k} = \frac{\binom{30}{k+1} \left(\frac{1}{4}\right)^{k+1} \left(\frac{3}{4}\right)^{29-k}}{\binom{30}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{30-k}}$$

$$= \frac{30!}{(k+1)!(29-k)!} \times \frac{1}{4}$$

$$= \frac{30!}{k!(30-k)!} \times \frac{3}{4}$$

$$= \frac{1}{3} \frac{(30-k)}{k+1}$$

$$= \frac{30-k}{3k+3}$$

We need $P_{k+1} > P_k$

$$\therefore \frac{P_{k+1}}{P_k} > 1$$

$$\frac{30-k}{3k+3} > 1$$

$$30-k > 3k+3$$

$$27 > 4k$$

$$k < \frac{27}{4}$$

$$\therefore k = 6$$

$$P_7 = \binom{30}{7} \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^{23}$$

$$= 0.166235674 \rightarrow$$

$$\approx 16.6\%$$

$\therefore$ 16.6% chance he guesses 7 correctly.

Question 13

(d) $\frac{dT}{dt} = -k(T-20)$

$$\begin{aligned} \text{(i)} \quad T &= 20 + Ae^{-kt} \\ \frac{dT}{dt} &= -kAe^{-kt} \\ &= -k(Ae^{-kt} + 20 - 20) \\ &= -k(T-20) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \text{When } t=0, T=160. \\ 160 &= 20 + Ae^0 \\ \therefore A &= 140 \\ 100 &= 20 + 140e^{-10k} \\ 80 &= 140e^{-10k} \\ \ln\left(\frac{80}{140}\right) &= -10k \\ k &= \frac{\ln\left(\frac{4}{7}\right)}{-10} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad 50 &= 20 + 140e^{\frac{\ln\left(\frac{4}{7}\right)t}{10}} \\ \ln\left(\frac{3}{14}\right) &= \frac{\ln\left(\frac{4}{7}\right)}{10} t \\ t &= \frac{10 \ln\left(\frac{3}{14}\right)}{\ln\left(\frac{4}{7}\right)} \\ &= 27.52683313... \\ &\approx 28 \text{ minutes.} \end{aligned}$$

Question 14

a) (i) $x = \frac{2}{\sqrt{3}} \cos 3t + 2 \sin 3t$

$$\dot{x} = \frac{-2}{\sqrt{3}} \sin 3t + 6 \cos 3t$$

$$\begin{aligned} \ddot{x} &= \frac{-18}{\sqrt{3}} \cos 3t - 18 \sin 3t \\ &= -9\left(\frac{2}{\sqrt{3}} \cos 3t + 2 \sin 3t\right) \\ &= -3^2 x \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad R^2 &= a^2 + b^2 & \tan \theta &= \frac{\frac{2}{\sqrt{3}}}{2} \\ &= \sqrt{\frac{4}{3} + 4} & &= \frac{1}{\sqrt{3}} \\ &= \frac{\sqrt{16}}{\sqrt{3}} & \therefore \theta &= \frac{\pi}{6} \\ &= \frac{4}{\sqrt{3}} \end{aligned}$$

$$\therefore x = \frac{4}{\sqrt{3}} \sin\left(3t + \frac{\pi}{6}\right)$$

$$\therefore \text{period} = \frac{2\pi}{3}, \text{ amplitude} = \frac{4}{\sqrt{3}}$$

b) $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$

Integrating both sides w.r.t. x .

$$\frac{(1+x)^{n+1}}{n+1} = \sum_{k=0}^n \binom{n}{k} \frac{x^{k+1}}{k+1} + C$$

Let $x=0$:

$$\begin{aligned} \frac{1}{n+1} &= 0 + C \\ C &= \frac{1}{n+1} \end{aligned}$$

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(b) continued.

Let $x=1$:

$$\frac{2^{n+1}}{n+1} = \sum_{k=0}^n \frac{1^{k+1}}{k+1} \binom{n}{k} + \frac{1}{n+1}$$

$$\frac{2^{n+1}-1}{n+1} = \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k}$$

(c) (i) $\ddot{y} = -g$

When $t=0$, $\dot{y} = V \sin \theta$.

$$\therefore C = V \sin \theta$$

$$\dot{y} = \frac{-gt^2}{2} + Vt \sin \theta + C$$

When $t=0$, $y=h$

$$h = 0 + 0 + C$$

$$\therefore C = h$$

$$\therefore y = \frac{-gt^2}{2} + Vt \sin \theta + h$$

(ii) $t = \frac{x}{V \cos \theta}$

$$\begin{aligned} y &= V \left(\frac{x}{V \cos \theta} \right) \sin \theta - \frac{g}{2} \left(\frac{x^2}{V^2 \cos^2 \theta} \right) + h \\ &= h + x \tan \theta - \frac{g}{2V^2 \cos^2 \theta} x^2 \end{aligned}$$

(iii) Ball clears net if $y > h$ when $x=9$.

$$h + 9 \tan \theta - \frac{81g}{2 \cos^2 \theta \cdot V^2} > h$$

$$9 \tan \theta > \frac{81g}{2 \cos^2 \theta \cdot V^2}$$

$$V^2 > \frac{9g}{2 \cos^2 \theta \tan \theta}$$

$$V^2 > \frac{9g}{2 \sin \theta \cos \theta}$$

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c) (iv) Ball hits attack line when $x=12$, $y=0$ and $v^2 > \frac{9g}{2 \sin \theta \cos \theta}$

Now,

$$0 = h + 12 \tan \theta - \frac{144g}{2V^2 \cos^2 \theta}$$

$$\frac{144g}{2V^2 \cos^2 \theta} = h + 12 \tan \theta$$

$$\frac{144g}{2 \cos^2 \theta} = (h + 12 \tan \theta) V^2$$

$$\frac{144g}{2 \cos^2 \theta} > \frac{(h + 12 \tan \theta) 9g}{2 \sin \theta \cos \theta}$$

$$\frac{8}{\cos^2 \theta} > \frac{h + 12 \tan \theta}{2 \sin \theta}$$

$$16 \tan \theta > h + 12 \tan \theta$$

$$4 \tan \theta > h$$

$$\tan \theta > \frac{h}{4}$$

(v) $\tan \theta > \frac{h}{4}$

$$\tan \theta > \frac{2.43}{4}$$

$$\theta > \tan^{-1}\left(\frac{2.43}{4}\right)$$

$$\theta > 31.28^\circ$$